



WORKING IN RESEARCH

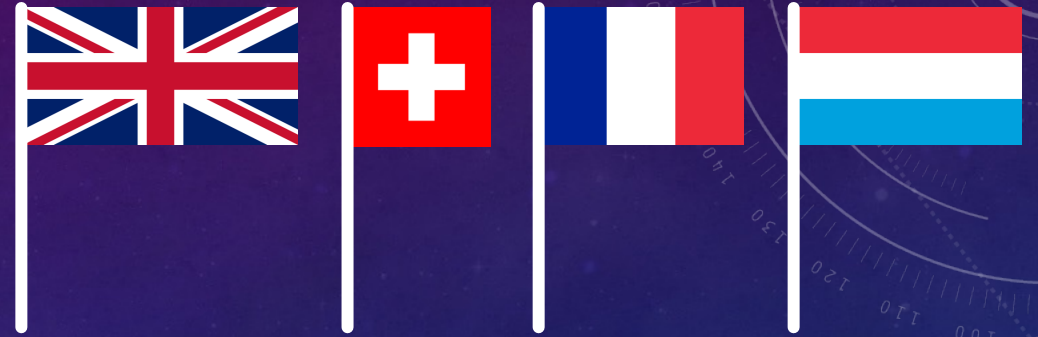
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LUXEMBOURG

(VERY) QUICK INTRODUCTION



- I was born in Burnley and I'm 26.
- I speak English and French.
- I have lived in the UK, Switzerland, France and Luxembourg.
- I'm a **researcher** at a research group called **SnT**, in **Luxembourg**.
 - I research things that fit into what I would call Mathematical Computer Science.
- I did my PhD at CERN in Geneva, Switzerland.
- In general, I love maths, computer science, and combining them to do wonderful things 😊

MY PATH SO FAR

- **A-Levels at NCC** – Maths, Computing and Physics.
- **Bachelors degree** (with a year at CERN) in Computer Science and Mathematics at **Manchester**.
- Whilst keeping my Manchester affiliation, I did my **PhD in Computer Science** at **CERN in Geneva**, Switzerland.
- Now, researcher in Luxembourg + external consultant with CERN.

NELSON AND COLNE
COLLEGE

MANCHESTER
1824
The University of Manchester




UNIVERSITÉ DU
LUXEMBOURG

THE THINGS I DON'T PUT ON MY CV

- A-Levels did not come naturally to me and I had to work really hard!
- At NCC, the only subject in which I had some of the best grades was Computing.
 - Because I'd been programming since I was 13.
- Got a D in my first A-Level maths exam... Jo Turner worked miracles and I eventually got an A overall!
- I did an EPQ and got an E!
- I **only just** got into Manchester's bachelors degree programme.
- I arrived in Manchester expecting to just hang on, and never do very well.

BUT THEN...



Graduated from my bachelors at Manchester with an excellence award



Went to CERN when I was 20
Got a fully-funded PhD position at CERN when I was 22

Instrumentation

Very jetlagged...

Original file is kept but renamed to force imports to use the instrumented bytecode.

AST modification - automatic instrumentation

Compilation to Python bytecode

Enough information to check behaviour

an augmented approach and uses this analysis, which is conservative instrumentation.

Instrumentation is performed by adding ASTs of instrumentation code to the AST representation of the program, and then compiling to bytecode.



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MY FAVOURITE BITS SO FAR

- 4 years at CERN working with extremely talented people.
- Leading a small team at CERN to develop my PhD research.
- Seeing programs run for the first time that took months of theoretical development and software engineering. Knowing that you're the first person ever to do it is a special feeling 😊

International Conference on Computing in High Energy Physics
2019, Adelaide, Australia

MY FAVOURITE BITS SO FAR

- Giving talks around the world to describe research – I've been fortunate to give talks in the UK, Switzerland, the Czech Republic, Portugal and Australia.
- That list would have included the US if COVID hadn't happened ☹



cern **MANCHESTER**

VyPR2: A Framework for Runtime Verification of Python Web Services

cern.ch/vyp

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VyPR2 verifies Python-based web services with respect to specifications in Control-Flow Temporal Logic, a logic whose intention is to make describing state and time constraints of running programs easy for engineers.

CFTL Specification

"Every time x is assigned True , the next call to insert should take less than 1 unit of time."

Binding Space Construction

B_φ A space of maps from variables to vertices in the SCFG that might generate a binding at runtime. Elements of this space are called static bindings.

PyCFTL code

Engineer

```
forall(q = changes('x')) \
  Check(lambda q : \
    If(q['x'] equals(True)) then { \
      q.next_call('insert').duration() <= (0, 1) \
    } \
  )
```

Symbolic Control-Flow Graph

VyPR

SCFG(P) = (V, E, v_s)
A statically computable, directed graph derived from a program P .
 V is a set of vertices which encode whether program variables/functions have been assigned/called. v_s is the starting vertex.

Atom-driven Instrumentation

VyPR

$A(\varphi)$ For each static binding in the binding space B_φ , we iterate through the atoms contained in the specification and uses them to find interesting points in the SCFG that might generate observations.

Efficient Monitoring at Runtime

VyPR

Every time a static binding is traversed at runtime, a concrete binding is generated. For each concrete binding, we instantiate an and-or tree. Lookup of which trees to update is efficient given VyPR's use of data from instrumentation.

Verdict Server

VyPR2

MY LEAST FAVOURITE BITS SO FAR

- The PhD was stressful!
- It feels good when you're the first person to do something.... but if it goes wrong.... you're the only person in the world who can fix it!
 - Not normally a problem, but tiredness can make it worse 😞
- For the 3 years of my PhD in Geneva, my partner had to live in the UK.
 - Approximately 60 flights in 3 years – I'm easyjet's biggest fan.

WHAT ABOUT THE RESEARCH?

- **In general, I work on developing mathematically-founded ways to analyse computer systems.**
- This involves using tools from maths such as graphs, sets, functions, (temporal) logic, differential and integral calculus.
- Problems are usually mixtures of many different areas – we never solve a problem and think “okay this is a differentiation problem”.

WHAT DO I RESEARCH?

How can we formally describe the behaviour of a computer system so that we can automatically check and perform diagnostics when something goes wrong?

- **Key problems**
 - Developing a mathematical formalism with which we can say what should be true during an execution of a computer system.
 - Deciding how to get the information we need out of a computer system in order to check whether things are true.
 - Deciding whether the information we get is what we expect.
 - If the information we got wasn't what we expected, working out why.
- Multiple areas of maths and computer science are needed.
- One problem could need many areas.

WHAT SKILLS DO I USE?

- Mathematics (or whatever the skill equivalent of maths is)
 - In my case maths A-Level, half-maths Bachelors, maths-based PhD
- Programming and Software Engineering
 - If writing the code to solve a problem is difficult, things will be slowed down.
 - I have to write a decent amount of code and design software.
- Public speaking
 - A key part of disseminating and developing research is being able to present it to peers.
- Writing
 - A key part of a researcher's job is writing up work and publishing, as a way to disseminate knowledge.

WHY I LEFT THE UK

- I never intended to leave the UK...
- But I got a 1-year position at CERN when I was 20, so I got to give it a try!
- After that 1 year in Switzerland, I didn't want to go back.
- Moving country can be daunting!
 - Admin, new culture, new language

THE SMALL MATTER OF NEXT STEPS

- There is a lot of stress around deciding what to do next.
- I can give my opinion on the university path.
- Your choice should be **yours**, not what the people around you tell you is the best option.
- **The most famous university isn't necessarily the one for you.**
- **If you decide to go into research, the place you did your bachelors matters less and less the further away from your bachelors you get.**
 - In my case, pretty much no one cares (either what I studied or where I studied it).
- Academic prestige is not the only factor – lifestyle, surroundings, language, etc...
- My biggest regret from my bachelors: **I worked too hard.** Find somewhere where the balance is right for you.

TAKING CARE OF YOURSELF

Success is not worth burning out for.

- Achievement is part of your life, not all of it 😊
- Other aspects are just as important: exercise, eating properly, socialising (maybe not right now...).

In general, it shouldn't be hard to answer the question "What do you do for fun?"

A BIT OF A PARADOX

- During my PhD, I would often think “but I’m enjoying it – why is it a problem if I work 12 hours in a day?”
- Even if you enjoy something, your brain needs a break from it 😊

IF YOU HAVE NO IDEA WHAT TO DO NEXT?

- I was very lucky – I realised very young that I love Computer Science. Maths – not so much.
- I heavily disliked Maths until I was around 18 (thanks Jo!).
- **I took the Maths A-Level because it was a prerequisite for the courses at University that I wanted to do.**
- You may be the same! You may not have realised yet what you're really passionate about.
- To find the thing that makes you want to give up your free time to work.... can take a while 😊

SO TO RECAP

- Find what you love – **you will spend 40 hours of your week in work...** make sure you enjoy it 😊
- Explore – **find where you like to live, learn a new language, do things that aren't related to your studies**
- Remember that work/studying is a **part of** your life. You're allowed to do other things.

ANY QUESTIONS? 😊